Multivariate Random Variables:

$$X = (X_1, X_2, \dots, X_n)$$

All discrete: Joint PMF: $P_{X_1 \dots X_n}(x_1, \dots x_n)$ or $P_X(x)$

$$= P(X_1 = x_1, X_2 = x_2, \dots X_n = x_n)$$

$X_i \in \mathcal{X}_i$; $x_i \in \mathcal{X}_i$ $\quad \mathcal{X}_1 \to$ discrete set

$(x_1, x_2, \dots, x_n) \in \mathcal{X}_1 \times \mathcal{X}_2 \dots \times \mathcal{X}_n$

$S \subseteq [n] \triangleq \{1, 2, \dots, n\}$

$X_S = (X_i : i \in S)$ sub-vector

$x_S = (x_i : i \in S)$

$$P_{X_S}(x_S) = \sum_{x_{S^c}} P_X(x)$$

$F_{X_1 X_2 \dots X_n}(x_1, x_2 \dots x_n) \triangleq \{\omega \mid X_1(\omega) \le x_1, X_2(\omega) \le x_2 \dots X_n(\omega) \le x_n\}$

multivariate joint CDF

Special cases:

(i) $X_1, X_2, \dots X_n \to$ discrete

joint PMF $P_{X_1 X_2 \dots X_n}(X_1 = a_i, X_2 = b_2 \dots)$

(ii) jointly continuous: $X_1, X_2 \dots X_n$ are jointly continuous if the joint CDF can be expressed as

$$F_{X_1 X_2 \dots X_n}(x_1, x_2 \dots x_n) = \int_{-\infty}^{x_1} \int_{-\infty}^{x_2} \dots \int_{-\infty}^{x_n} f_{X_1 X_2 \dots X_n}(\tau_1, \tau_2, \dots \tau_n)\, d\tau_n \dots d\tau_2\, d\tau_1$$

$f_{X_1 X_2 \dots X_n}$: joint PDF

Independance:

$X_1, X_2, \dots X_n$ are said to be independent if for every $C_1, C_2, \dots C_n \subseteq \mathbb{R}$

$$P(X_1 \in C_1, X_2 \in C_2 \dots X_n \in C_n) = \prod_{i=1}^{n} P(X_i \in C_i)$$

for "nice" sets C_i

→ $P(X_1 \in C_i)$ is well-defined

in particular taking $C_i = (-\infty, x_i]$

$$\Rightarrow F_{X_1 X_2 \dots X_n}(x_1, x_2 \dots x_n) = \prod_{i=1}^{n} F_{X_i}(x_i)$$

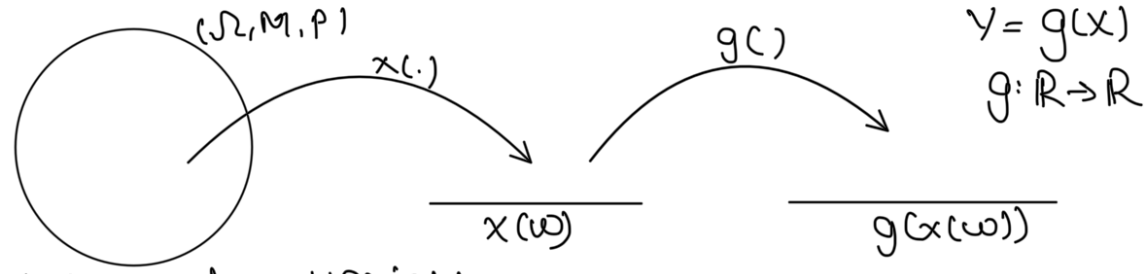
Conditional independance: X and Y are said to be conditionally independent given Z if

$$F_{X,Y|Z}(x,y|z) = F_{X|Z}(x|z) \cdot F_{Y|Z}(y|z)$$

$$P(X \le x, Y \le y | Z = z) = P(X \le x | Z = z) \cdot P(Y \le y | Z = z) \quad \forall\, x,y,z \in \mathbb{R}$$

if true: $\rightarrow$ $X \rightarrow Z \rightarrow Y$ forms a Markov Chain

Functions of Random variables



$X \rightarrow$ random variable

$Y = g(X)$ if $X \rightarrow$ discrete $\Rightarrow$ $Y \rightarrow$ discrete

$X \rightarrow$ PMF $P_X(\cdot) \Rightarrow P(Y=y) = \sum_{i:g(x_i)=y} P_X(x_i)$

$Y = g(X)$ $X \sim$ continuous with pdf $f_X(\cdot)$

$X \sim$ uniform $[-1,1]$ $\quad Y = X^2$

$$F_Y(y) = P(Y \le y) = P(X^2 \le y) = P(-\sqrt{y} \le y \le \sqrt{y}) = \sqrt{y} \rightarrow f_Y(y) = \frac{1}{2\sqrt{y}} \quad y \in (0,1]$$

Theorm: Suppose X is a R.V with a continuous CDF $F_X(\cdot)$. Then define $U = F_X(x)$. Then $U \sim$ uniform $[0,1]$

Proof: $P(U \le u) = P(F_X(x) \le u) = P(x \le F_X^{-1}(u)) = P(X \le F_X^{-1}(u))$

$= F_X(F_X^{-1}(u)) = u \quad u \in [0,1]$

Corollary: If $U \sim$ uniform $[0,1)$. $F_X(\cdot)$ is any continuous CDF then $F_X^{-1}(U)$ has CDF $F_X(\cdot)$

$Y = g(X)$ $\quad X \sim$ pdf $f_X(\cdot)$

Suppose $g(\cdot)$ is a one-to-one and differentiable then

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

$Y = \max(x_1, x_2 \ldots x_n)$ $\quad Z = \min(x_1, x_2, \ldots x_n)$

joint CDF $= F_{X_1 X_2 \ldots X_n}(x_1, x_2, \ldots x_n)$

$F_U(u) = P(\max(x_1, x_2 \ldots x_n) \le u)$

$F_Z(z) = P(Z \geq z)$

$\Rightarrow 1 - F_Z(z) = P(X_1 \geq z \dots X_n > z)$

$= P(X_1 \leq y, X_2 \leq y, \dots X_n \leq y)$

$= F_{X_1 X_2 \dots X_n}(y, y, y \dots y)$

$X \sim \exp(\lambda)$ $Y \sim \exp(\mu)$ Independent

$Z = \min(X, Y)$ $P(Z > z) = P(\min(X,Y) \geq z) = P(X > z) \cdot P(Y > z)$

$\Rightarrow Z \sim \exp(\lambda + \mu)$ $= e^{-\lambda z} \cdot e^{-\mu z} = e^{-(\lambda+\mu)z}$

X & Y → discrete RV's ; joint PMF $P_{XY}(\cdot,\cdot)$. $Z = X + Y$

$$P(Z = K) = \sum_{l} P(X = l, Y = K - l) = \sum_{l} P_{X,Y}(l, K-l)$$

If X, Y → independent $\Rightarrow P_Z = P_X * P_Y$

let X, Y → jointly continuous with joint PDF $f_{X,Y}(\cdot,\cdot)$

$Z = X + Y$

$$F_Z(z) = P(Z \leq z) = P(X + Y \leq z) = \iint_{x+y \leq z} f_{XY}(x, y)\, dx\, dy$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_{XY}(x, y)\, dy\, dx \qquad y = \tau - x$$

$$\Rightarrow \int_{-\infty}^{z} \underbrace{\int_{-\infty}^{\infty} f_{X,Y}(x, \tau - x)\, dx}_{f_Z(\tau)}\, d\tau$$

If X, Y are independent $\Rightarrow f_Z(t) = f_X * f_Y = \int_{-\infty}^{\infty} f_X(x)\, f_Y(t - x)\, dx$

$Y_1 = g_1(X_1 \dots X_n)$
$\vdots$
$Y_n = g_n(X_1 \dots X_n)$
} one-one & all g_i's are differentiable

Given joint PDF $f_{X_1 X_2 \dots X_n}(x_1, \dots x_n)$

$$J = \begin{pmatrix} \frac{dx_1}{dy_1} & \dots & \frac{dx_n}{dy_1} \\ \vdots & & \vdots \\ \frac{dx_1}{dy_n} & & \frac{dx_n}{dy_n} \end{pmatrix}$$

$f_{Y_1 Y_2 \dots Y_n}(y_1, \dots y_n) = f_{X_1 \dots X_n}(S_1(y_1 \dots y_n), S_2(y_2 \dots y_n) \dots)$

$x_1 = S_1(y_1, \dots y_n) \dots x_n = S_n(y_1, \dots y_n)$

Discrete: $Y = g(X)$

$$P_Y(y) = \sum_{x: g(x)=y} P_X(x)$$

Continuous:

→ CDF method:

$$F_Y(y) = P(Y \le y) = P(X \in \{x: g(x) \le y\})$$
$$f_Y(y) = F_Y'(y)$$

→ PDF method:

Solve $y = g(x)$ to get $x_1(y), x_2(y) \dots$

$$f_Y(y) = \sum_{\substack{i=1,2,\dots \\ x_i(y) \in \text{supp}(x)}} f_X(x_i(y)) \cdot \frac{1}{|g'(x_i(y))|}$$

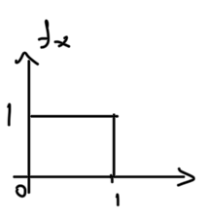
$= |(g^{-1})'(y)|$

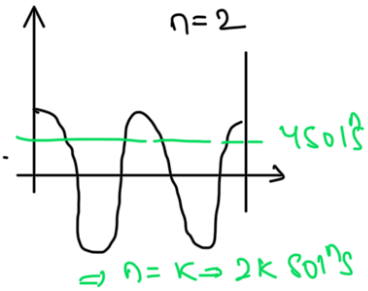
g is monotonic ⇒ only 1 term

$$f_X(x) \to \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \quad \forall x \qquad (\text{mean} = 0, \text{SD} = 1)$$

$$f_Y(y) \to \frac{1}{\sqrt{2\pi}|a|} e^{-(y-b)^2/2a^2} \quad \forall y \qquad (\text{mean} = b, \text{SD} = |a|)$$

$Y = \cos(2\pi n X) \quad n = 1, 2, \dots$





$$f_Y(y) = \frac{1}{2\pi n \sqrt{1-y^2}} \cdot 2n = \frac{1}{\pi\sqrt{1-y^2}} \quad (-1 < y < 1)$$

independent of n

2 variables → 2 variables

$X, Y \sim f_{XY}$

$(x, y) \to (u, v)$ $\quad u(x,y)$, $v(x,y)$ $\quad$ inverse ↓ $x(u,v)$, $y(u,v)$

$$du\,dv = |\|J\||\,dx\,dy$$

$$J = \begin{bmatrix} \frac{du}{dx} & \frac{du}{dy} \\ \frac{dv}{dx} & \frac{dv}{dy} \end{bmatrix}$$

$$f_{uv}(u,v) = f_{xy}(x(u,v), y(u,v)) \cdot \frac{1}{|\|J\||}$$

$$\begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{bmatrix}$$

CDF → $(U \le u ; V \le v) \longleftrightarrow$ suitable region in (x,y)

$$F_{UV}(u,v) = \iint_{\text{region}} f_{XY}(x,y)\,dx\,dy$$

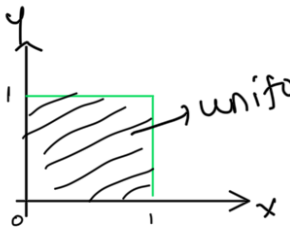
$$f_{UV}(u,v) = \frac{\partial^2}{\partial x \partial y} F_{U,V}$$

ⓐ $f_{XY}(x,y) = \frac{1}{2\pi} e^{-x^2/2} e^{-y^2/2} \quad \forall x,y$

$U = ax+by$
$V = cx+dy$

$$f_{UV}(u,v) = \frac{1}{2\pi(ad-bc)} e^{\frac{-(du-bv)^2 + (-cu+av)^2}{2(ad-bc)^2}}$$

$\forall u,v$

ⓑ 

$U = x+y$
$V = x-y$

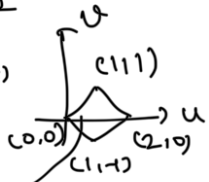
$x = \frac{u+v}{2}$
$y = \frac{u-v}{2}$

$|J| = 2$

$0 < x < 1$
$0 < y < 1$

$0 < \frac{u+v}{2} < 1$
$0 < \frac{u-v}{2} < 1$

$0 < u+v < 2$
$0 < u-v < 2$ ⇒



$$f_{UV}(u,v) = \frac{1}{2} \cdot I_{(u,v)}$$

ⓒ $U = \max(x,y) \quad V = \min(x,y)$

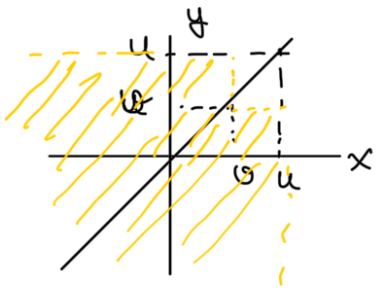
$\text{supp}(x,y) \to \mathbb{R}^2 \qquad \text{supp}(u,v) \to u > v$

$$f_{UV}(u,v) = f_{XY}(u,v) \cdot 1 + f_{XY}(v,u) \cdot 1$$

CDF method ⇒ $F_{UV}(u,v) = P(U \le u, V \le v)$

$= P(\max(x,y) \le u, \min(x,y) \le v)$

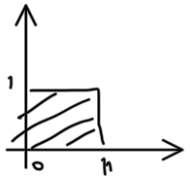
$= F_{XY}(u,v) + F_{XY}(v,u) - F_{XY}(v,v)$



(d) $u = x+y \quad v = y \quad x, y \sim$ iid Uniform $[0,1]$

$$f_u(u) = \int_{v=0}^{1} f_{xy}(u-v, v)\, dv \qquad u \in [0,2]$$

$$= \int_0^1 f_x(u-v) f_x(v)\, dv$$



↓ convolution

(e) $u = x/y \quad v = y \quad x, y \sim$ iid $\frac{1}{\sqrt{2\pi}} e^{-x^2/2}$ all x

$$f_u(u) = \int_{v=-\infty}^{\infty} \frac{1}{2\pi} e^{-(u^2v^2+v^2)/2} |v|\, dv = \frac{1}{\pi(1+u^2)}$$

$u \in \mathbb{R}$

Cauchy distribution

Multivariate distributions:

$x_1, x_2, \ldots x_n \rightarrow$ jointly distributed

all discrete $\quad p_{x_1 x_2 \ldots x_n}(x_1, x_2 \ldots x_n)$

$$S \subseteq [n] \triangleq \{1, 2, \ldots n\}$$

$$p_{x_S}(x_S) = \sum_{x \in x_{S^c}} p_{x_1 x_2 \ldots x_n}(x_1, x_2 \ldots x_n)$$

jointly continuous:

$$f_{x_S}(x_S) = \int_{x_{S^c}} \cdots \int f_{x_1 x_2 \ldots x_n}(x_1, x_2 \ldots x_n)\, dx_1 \ldots dx_n$$

$$S, T \subseteq [n] \qquad S \cap T = \emptyset$$

$$p_{x_S / x_T = x_T}(x_S) = \frac{p_{x_{SUT}}(x_{SUT})}{p_{x_T}(x_T)}$$

Examples:

① $x_1, x_2, \ldots x_n \sim$ iid Bernoulli(p)

$$Y = x_1 + x_2 + \ldots + x_n \sim \text{Bernoulli}(n, p)$$

② $x_1, x_2, \ldots x_n \sim$ iid geo(p)

$x_i \in \{1, 2, 3, \dots\}$

$\downarrow \quad \downarrow \quad \searrow$

$p \quad (1-p)p \quad (1-p)^2 p$

$S_2 = x_1 + x_2 \in \{2, 3, \dots\}$

$P(S_2 = m) = mp^2(1-p)^{m-2}$

$S_k = x_1 + x_2 + \dots + x_k \in \{k, k+1, \dots\}$

$P(S_k = m) = {}^{m-1}C_{k-1}\, p^k (1-p)^{m-k}$